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United States Patent	12407077
Kind Code	B2
Date of Patent	September 02, 2025
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Multi-band filter

Abstract

A multi-band filter includes a circuit board, a first resonator, a second resonator, and a coupling element configured to couple the first resonator with the second resonator. The coupling element includes a first coupling capacitor, a second coupling capacitor, a first short-circuited stub and a second short-circuited stub. The first coupling capacitor has two terminals electrically connected to the first portion of the first resonator and the first portion of the second resonator respectively. The second coupling capacitor has two terminals electrically connected to the second portion of the first resonator and the second portion of the second resonator respectively. The first short-circuited stub is electrically connected to the first coupling capacitor and a ground plane. The second short-circuited stub is electrically connected to the second coupling capacitor and a ground plane.

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Appl. No.: 18/338322

Filed: June 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240097300 A1	Mar. 21, 2024

Foreign Application Priority Data

TW	111134823	Sep. 15, 2022
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Publication Classification

Int. Cl.: H01P1/203 (20060101); H01P1/205 (20060101); H01P1/213 (20060101)

U.S. Cl.:

CPC H01P1/2135 (20130101); H01P1/203 (20130101); H01P1/20381 (20130101); H01P1/205 (20130101);

Field of Classification Search

CPC: H01P (1/203); H01P (1/20); H01P (1/201); H01P (1/20327); H01P (1/20336); H01P (1/20345); H01P (1/205); H01P (1/2135); H01P (1/20381)

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Taiwan Application Ser. No. 111134823, filed Sep. 15, 2022, which is herein incorporated by reference.

BACKGROUND

Field of Invention

(2) The present invention relates to a multi-band filter.

Description of Related Art

(3) With the rapid advancement of technology in recent years, electronic products such as PCs, tablet PCs, NBs, and smartphones have become indispensable in our daily lives. These devices incorporate various radio frequencies to cater to the diverse demands of users, and filters are commonly used to eliminate unwanted signal components within specific frequency bands. However, when it comes to the millimeter wave frequency band, traditional filters suffer from higher signal loss and require larger circuit areas. Consequently, there is a pressing need for a new type of filter that can effectively address these challenges and offer improved performance in millimeter wave frequency bands.

SUMMARY

(4) Embodiments of the present invention provide a multi-band filter capable of filtering and

outputting plural frequency bands. The multi-band filter in the embodiments of the present invention has less loss and also has a small circuit area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows.
- (2) FIG. 1 is a schematic diagram showing a structure of a multi-band filter in accordance with an embodiment of the present invention.
- (3) FIG. 2 is a schematic diagram showing a structure of a multi-layer circuit board in accordance with an embodiment of present invention.
- (4) FIGS. 3-5 are schematic diagram showing a structure of a multi-band filter disposed in the multi-layer circuit board.
- (5) FIG. 6 is a schematic diagram showing a structure of an antenna module in accordance with an embodiment of the present invention.
- (6) FIG. 7 is a schematic diagram showing a structure of an antenna module after being bent in accordance with an embodiment of the present invention.
- (7) FIG. 8 is a schematic diagram showing a virtual folded line of the multi-band filter in accordance with an embodiment of the present invention.
- (8) FIG. 9 is a schematic diagram showing another virtual folded line of the multi-band filter in accordance with an embodiment of the present invention.
- (9) FIG. 10 is a schematic diagram showing another virtual folded line of the multi-band filter in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

- (10) Specific embodiments of the present invention are further described in detail below with reference to the accompanying drawings, however, the embodiments described are not intended to limit the present invention and it is not intended for the description of operation to limit the order of implementation. Moreover, any device with equivalent functions that is produced from a structure formed by a recombination of elements shall fall within the scope of the present invention. Additionally, the drawings are only illustrative and are not drawn to actual size.
- (11) The using of “first”, “second”, “third”, etc. in the specification should be understood for identifying units or data described by the same terminology but are not referred to particular order or sequence. FIG. 1 is a schematic diagram showing a structure of a multi-band filter **100** in accordance with an embodiment of the present invention. The multi-band filter **100** is disposed in a circuit layer of a circuit board to provide plural frequency bands. In this embodiment, the multi-band filter **100** is disposed on a substrate made of liquid crystal polymer (LCP), but other embodiments of the present invention are not limited thereto. For example, polyimide (PI) or modified PI (MPI) is common material used for the substrate. The multi-band filter **100** includes a first resonator **110**, a second resonator **120** and a coupling element located therebetween. The coupling element includes a first coupling capacitor **131**, a second coupling capacitor **132**, a first short-circuited stub **141** and a second short-circuited stub **142**. I/O terminals **101** and **102** of the multi-band filter **100** are respectively located at the first resonator **110** and the second resonator **120** to receive/output an electric signal. For example, an electric signal desired to be filtered may be inputted to the multi-band filter **100** through the I/O terminal **IO1**, and then a filtered electric signal is outputted through the I/O terminal **IO2**. Vice versa, the electric signal desired to be filtered may be inputted to the multi-band filter **100** through the I/O terminal **102**, and then the filtered electric signal is outputted through the I/O terminal **101**. In this embodiment, a zero-degree feed structure is applied for the electric signal to improve performance of stop bands, but other embodiments of the

present invention are not limited thereto. In this embodiment, the electric signal is in millimeter wave range, and the multi-band filter **100** provides 28 GHz and 39 GHz frequency bands simultaneously. For example a frequency band from 26.13 GHz to 28.55 GHz and a frequency band from 37.56 GHz to 41.25 GHz. However, other embodiments of the present invention are not limited thereto.

(12) The first resonator **110** has a first portion **110a** and a second portion **110b** opposite to each other. The first portion **110a** of the first resonator **110** is electrically connected to the first coupling capacitor **131**, and the second portion **110b** of the first resonator **110** is electrically connected to the second coupling capacitor **132**. The second resonator **120** has a first portion **120a** and a second portion **120b** opposite to each other. The first portion **120a** of the second resonator **120** is electrically connected to the first coupling capacitor **131**, and the second portion **120b** of the second resonator **120** is electrically connected to the second coupling capacitor **132**.

(13) In order to match a high frequency band, the first resonator **110** is electrically connected to the first coupling capacitor **131** and the second coupling capacitor **132** through a first funnel-shaped connecting path **151** and a second funnel-shaped connecting path **152**. The second resonator **120** is electrically connected to the first coupling capacitor **131** and the second coupling capacitor **132** through a third funnel-shaped connecting path **153** and a fourth funnel-shaped connecting path **154**. The first funnel-shaped connecting path **151** is disposed between the first portion **110a** of the first resonator **110** and the first coupling capacitor **131** to provide electric connection therebetween. The second funnel-shaped connecting path **152** is disposed between the second portion **110b** of the first resonator **110** and the second coupling capacitor **132** to provide electric connection therebetween. The third funnel-shaped connecting path **153** is disposed between the first portion **120a** of the second resonator **120** and the first coupling capacitor **131** to provide electric connection between the second resonator **120** and the first coupling capacitor **131**. The fourth funnel-shaped connecting path **154** is disposed between the second portion **120b** of the second resonator **120** and the second coupling capacitor **132** to provide electric connection therebetween.

(14) The first funnel-shaped connecting path **151** includes an escalating path. As shown in FIG. 1, the wider path portion is directly connected to the first portion **110a** of the first resonator **110**, and the narrower path portion is directly connected to the first coupling capacitor **131**. Similarly, the second funnel-shaped connecting path **152** includes an escalating path. As shown in FIG. 1, the wider path portion is directly connected to the second portion **110b** of the first resonator **110**, and the narrower path portion is directly connected to the second coupling capacitor **132**. Similarly, the third funnel-shaped connecting path **153** includes an escalating path. As shown in FIG. 1, the wider path portion is directly connected to the first portion **120a** of the second resonator **120**, and the narrower path portion is directly connected to the first coupling capacitor **131**. Similarly, the fourth funnel-shaped connecting path **154** includes an escalating path. As shown in FIG. 1, the wider path portion is directly connected to the second portion **120b** of the second resonator **120**, and the narrower path portion is directly connected to the second coupling capacitor **132**.

(15) In some embodiments, the first resonator **110** and the second resonator **120** are half-wave resonators.

(16) In the embodiments of the present invention, the first coupling capacitor **131** and the second coupling capacitor **132** are interdigital capacitors respectively having a plurality of fingers. In this embodiment, the first coupling capacitor **131** and the second coupling capacitor **132** respectively have six fingers, and a length of each of the fingers is substantially equal to 135 μm , but embodiments of the present invention are not limited thereto. In other embodiments of the present invention, the number of the fingers and the length of each fingers may be adjusted in accordance with actual demands to adjust bandwidths of passbands of the multi-band filter **100**.

(17) The first short-circuited stub **141** and the second short-circuited stub **142** are electrically connected to the first coupling capacitor **131** and the second coupling capacitor **132**. The lengths of the first short-circuited stub **141** and the second short-circuited stub **142** may be adjusted to adjust a

bandwidth of a high frequency band. In this embodiment, the first short-circuited stub **141** and the second short-circuited stub **142** are designed to have plural bending portions and a short-circuited pattern directly connected thereto to decrease the planar area occupied by the multi-band filter **100**. For example, the first short-circuited stub **141** includes bending portions **141a** and **141b** and a short-circuited pattern **141c**, where the short-circuited pattern **141c** is electrically grounded through vias VG1. The second short-circuited stub **142** includes bending portions **142a** and **142b** and a short-circuited pattern **142c**, in which the short-circuited pattern **142c** is electrically grounded through vias VG2.

(18) In some embodiments of the present invention, the multi-band filter **100** may be divided into several filter portions and disposed in different layers in a multi-layer circuit board for decreasing surface area occupied by the multi-band filter **100**.

(19) In some embodiments, U-shaped grounded patterns **161**, **162** are arranged for the convenience of testing, in which the I/O terminals **101**, **102** are respectively surrounded by the U-shaped grounded patterns **161**, **162**. When a testing is performed, a testing equipment is electrically connected to the I/O terminals **101/102** and corresponding U-shaped grounded pattern **161/162** to enable signal conduction for performing the testing operation. For example, the I/O terminals **101/102** are grounded to perform tests and acquire parameters of the multi-band filter **100**. The elongated U-shaped patterns **161**, **162** allow multiple testing points, thereby decreasing deviation occurred when performing the testing operation on a single point. In this embodiment, the U-shaped grounded patterns **161**, **162** are grounded through vias VG3 and VG4 respectively.

(20) Referring to FIGS. 2-5, FIG. 2 is a schematic diagram showing a structure of a multi-layer circuit board **200** in accordance with an embodiment of present invention, and FIGS. 3-5 are schematic diagram showing a structure of a multi-band filter disposed in the multi-layer circuit board **200**. As shown in FIG. 2, the multi-layer circuit board **200** includes at least three layers. In this embodiment, the multi-layer circuit board **200** includes a first circuit layer **210**, a second circuit layer **220**, a third circuit layer **230**, a first insulation layer LC1 and a second insulation layer LC2. In the multi-layer circuit board **200**, the second circuit layer **220** is disposed on the first circuit layer **210**, and the third circuit layer **230** is disposed on the second circuit layer **220**, and thus the second circuit layer **220** is disposed between the first circuit layer **210** and the third circuit layer **230**. The first insulation layer LC1 is disposed between the first circuit layer **210** and the second circuit layer **220** to provide electrical insulation between the first circuit layer **210** and the second circuit layer **220**. The second insulation layer LC2 is disposed between the second circuit layer **220** and the third circuit layer **230** to provide electrical insulation between the second circuit layer **220** and the third circuit layer **230**. In this embodiment, the multi-layer circuit board **200** is a flexible circuit board, and the first insulation layer LC1 and the second insulation layer LC2 are made from liquid crystal polymer (LCP), but embodiments of the present invention are not limited thereto. For example, polyimide (PI) or modified PI (MPI) is common material used for the insulation material.

(21) The multi-band filter **100** is divided into two portions **100A** and **100B** respectively disposed in the first circuit layer **210** and the third circuit layer **230**, as shown in FIG. 3 and FIG. 5. The second circuit layer **220** includes a grounded layer **400** to provide a common ground plane for the filter portions **100A** and **100B**.

(22) Referring to FIG. 3, the filter portion **100A** is disposed in the first circuit layer **210** and includes a first portion **111** of the first resonator **110** (the first resonator **110** shown in FIG. 1), a first portion **121** of the second resonator **120** (the second resonator **120** shown in FIG. 1), the first funnel-shaped connecting path **151**, the third funnel-shaped connecting path **153**, the first coupling capacitor **131** and the first short-circuited stub **141**. In this embodiment, an area of the first portion **111** of the first resonator **110** is a half of an area of the first resonator **110**, and an area of the first portion **121** of the second resonator **120** is a half of an area of the second resonator **120**, but embodiments of the present invention are not limited thereto.

(23) Referring to FIG. 5, the filter portion **100B** is disposed in the third circuit layer **230** and

includes a second portion **112** of the first resonator **110** (the first resonator **110** shown in FIG. 1), a second portion **122** of the second resonator **120** (the second resonator **120** shown in FIG. 1), the funnel-shaped connecting path **152**, the fourth funnel-shaped connecting path **154**, the second coupling capacitor **132** and the second short-circuited stub **142**. In this embodiment, an area of the second portion **112** of the first resonator **110** is a half of an area of the first resonator **110**, and an area of the second portion **122** of the second resonator **120** is a half of an area of the second resonator **120**, but embodiments of the present invention are not limited thereto. In other embodiments of the present invention, other area ratios can be applied. For example, in one embodiment, the area of the first portion **111** of the first resonator **110** is one-third of the area of the first resonator **110**, and the area of the second portion **112** of the first resonator **110** is two-third of the area of the first resonator **110**. Similarly, the area of the first portion **121** of the second resonator **120** is one-third of the area of the second resonator **120**, and the area of the second portion **122** of the second resonator **120** is two-third of the area of the second resonator **120**.

(24) Referring to FIGS. 3-5 simultaneously, in order to electrically connect the filter portion **100A** with the filter portion **100B**, the filter portion **100A** located in the first circuit layer **210** and the filter portion **200B** in the third circuit layer **230** are partially overlapped, and thus vertical vias may be used to electrically connect the filter portion **100A** with the filter portion **100B**. Further, for convenience of grounding, also the grounded layer **400** is disposed between the filter portion **100A** located in the first circuit layer **210** and the filter portion **200B** in the third circuit layer **230**.

(25) For example, a via **V1** is disposed to vertically pass through the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulation layer **LC1** and the second insulation layer **LC2**, to electrically connect the first portion **111** of the first resonator **110** with the second portion **112** of the first resonator **110**. The via **V1** may pass through the grounded layer **400** in the second circuit layer **220**. To avoid that the via **V1** contacts the grounded layer **400** in the second circuit layer **220**, the second circuit layer **220** includes an insulation portion **410** surrounding the via **V1** in the second circuit layer **220** for electrical isolation between the via **V1** and the grounded layer **400**.

(26) For another example, a via **V2** is disposed to vertically pass through the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulation layer **LC1** and the second insulation layer **LC2**, to electrically connect the first portion **121** of the second resonator **120** with the second portion **122** of the second resonator **120**. The via **V2** may pass through the grounded layer **400** in the second circuit layer **220**. To avoid that the via **V2** contacts the grounded layer **400** in the second circuit layer **220**, the second circuit layer **220** includes an insulation portion **420** surrounding the via **V2** in the second circuit layer **220** for electrical isolation between the via **V2** and the grounded layer **400**.

(27) For still another example, vias **V3** are disposed to vertically pass through the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulation layer **LC1** and the second insulation layer **LC2**, to electrically connect the first short-circuited stub **141** with the second short-circuited stub **142**. The vias **V3** may pass through the grounded layer **400** in the second circuit layer **220**. Because it is necessary to ground the first short-circuited stub **141** and the second short-circuited stub **142**, vias **V3** are electrically connected to the grounded layer **400** in the second circuit layer **220** to achieve the grounding of the first short-circuited stub **141** and the second short-circuited stub **142**. In addition, vias **V4** are disposed to vertically pass through the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulation layer **LC1** and the second insulation layer **LC2**, to electrically connect the U-shaped grounded pattern **161** with the grounded layer **400** in the second circuit layer **220**. Vias **V5** are disposed to vertically pass through the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulation layer **LC1** and the second insulation layer **LC2**, to electrically connect the U-shaped grounded pattern **162** with the grounded layer **400** in the second circuit layer **220**.

(28) It can be understood from the above descriptions that the multi-band filter **100** is divided into

the filter portion **100A** and the filter portion **100B**, and the filter portion **100A** and the filter portion **100B** are disposed in different circuit layers of the multi-layer circuit board **200**. Through such design, a planar area occupied by the multi-band filter **100** can be significantly reduced. In addition, the lengths of the fingers of the first coupling capacitor **131** and the second coupling capacitor **132** may vary corresponding to the above design. As shown in FIG. 3, the lengths of the six fingers **131a** to **131f** of the first coupling capacitor **131** are respectively equal to 140 μm . Also, the lengths of the six fingers **132a**-**132f** of the second coupling capacitor **132** are respectively equal to 140 μm .

(29) Referring to FIG. 6, FIG. 6 is a schematic diagram showing a structure of an antenna module **600** in accordance with an embodiment of the present invention. The antenna module **600** includes a circuit board PCB, a radio frequency chip (RF IC) **610**, and antenna devices **620** and **630**. The circuit board PCB may be a single-layer circuit board including the multi-band filter **100** or a multi-layer circuit board (for example, the above multi-layer circuit board **200**) including the multi-band filter **100**. The radio frequency chip **610** and the antenna devices **620**, **630** are electrically connected to the multi-band filter **100** in the multi-layer circuit board **200**, to use the multi-band filter **100** to filter out unnecessary frequency bands. In this embodiment, technology of antenna in packaging (AiP) is used for packaging of the antenna module **600**. Comparing to conventional technology of system in package (SiP), the antenna packaging of this embodiment further integrate the antenna devices, and has a smaller size. For example, in this embodiment, the frequency chip **610** and the antenna devices **620**, **630** are disposed on the same surface of the multi-layer circuit board **200**. After the antenna module **600** is bent, such design may reduce the space occupied by the antenna module **600**, as shown in FIG. 7. After the antenna module **600** is bent, the frequency chip **610** and the antenna devices **620**, **630** are located on the inner side of the antenna module **600**, and thus the space occupied by the antenna module **600** is significantly decreased. In some embodiments, circuits of the circuit board can be used to form antenna devices, and thus the antenna devices **620**, **630** can be omitted and the space occupied by the antenna module **600** can be further decreased.

(30) Referring to FIG. 8, FIG. 8 is a schematic diagram showing a virtual folded line VFL1 of the multi-band filter **100** in accordance with an embodiment of the present invention. The virtual folded line VFL1 is configured to define a folding manner of the multi-band filter **100**, and thus the multi-band filter **100** can be bent in accordance with the virtual folded line VFL1. As shown in FIG. 8, the virtual folded line VFL1 is a straight line extended and passing through the first resonator **110** and the second resonator **120**, but the virtual folded line VFL1 does not pass through the first coupling capacitor **131** and the second coupling capacitor **132**. Therefore, when the multi-band filter **100** is folded (for example, the above case that the antenna module **600** is bent/folded), the multi-band filter **100** can be bent/folded along the virtual folded line VFL1 to satisfy a user's demand for a design of an antenna module. Because the virtual folded line VFL1 does not pass through the first coupling capacitor **131** and the second coupling capacitor **132**, the function of the multi-band filter **100** is not affected by the folding/bending of the multi-band filter **100**.

(31) Referring to FIG. 9, FIG. 9 is a schematic diagram showing a virtual folded line VFL2 of the multi-band filter **100** in accordance with an embodiment of the present invention. The virtual folded line VFL2 is configured to define another folding manner of the multi-band filter **100**. As shown in FIG. 9, the virtual folded line VFL2 is a straight line extending through the first resonator **110** and the second resonator **120**, but the virtual folded line VFL2 does not pass through the first coupling capacitor **131** and the second coupling capacitor **132**. Similar to the horizontal virtual folded line VFL1, the multi-band filter **100** can be bent/folded by using the folding manner defined by the virtual folded line VFL2 to satisfy a user's demand for a design of an antenna module. Further, because the virtual folded line VFL2 does not pass through the first coupling capacitor **131** and the second coupling capacitor **132**, the function of the multi-band filter **100** is not affected by the folding/bending of the multi-band filter **100**. In some embodiments, the diagonal virtual folded

line VFL2 can be arranged in another direction to define the folding manner, as shown in FIG. 10. (32) Although the present invention has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein. It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present invention without departing from the scope or spirit of the invention. In view of the foregoing, it is intended that the present invention cover modifications and variations of this invention provided they fall within the scope of the following claims.

Claims

1. A multi-band filter configured to provide a plurality of frequency bands, comprising: a circuit board; a first resonator having a first portion and an opposite second portion; a second resonator having a first portion and an opposite second portion; and a coupling element configured to couple the first resonator with the second resonator, wherein the coupling element comprises: a first coupling capacitor disposed between the first portion of the first resonator and the first portion of the second resonator, wherein a terminal of the first coupling capacitor is electrically connected to the first resonator, and another terminal of the first coupling capacitor is electrically connected to the second resonator; a second coupling capacitor disposed between the second portion of the first resonator and the second portion of the second resonator, wherein a terminal of the second coupling capacitor is electrically connected to the first resonator, and another terminal of the second coupling capacitor is electrically connected to the second resonator; a first short-circuited stub electrically connected to the first coupling capacitor and a ground plane; and a second short-circuited stub electrically connected to the second coupling capacitor and the ground plane; wherein the first resonator, the second resonator and the coupling element are disposed on the circuit board, and the first coupling capacitor and the second coupling capacitor are interdigital capacitors.

2. The multi-band filter of claim 1, wherein the circuit board is a multi-layer circuit board comprising at least three layers that at least one is grounded layer including the ground plane, and the first coupling capacitor and the second coupling capacitor are respectively disposed in different layers on opposite sides of the grounded layer.

3. The multi-band filter of claim 2, further comprises: a first funnel-shaped connecting path disposed between the first resonator and the first coupling capacitor, wherein the first funnel-shaped connecting path has an escalating width, the wider part of the first funnel-shaped connecting path is connected to the first resonator, the narrower part of the first funnel-shaped connecting path is connected to the first coupling capacitor; a second funnel-shaped connecting path disposed between the first resonator and the second coupling capacitor, wherein the second funnel-shaped connecting path has an escalating width, the wider part of the second funnel-shaped connecting path is connected to the first resonator, the narrower part of the second funnel-shaped connecting path is connected to the second coupling capacitor; a third funnel-shaped connecting path disposed between the second resonator and the first coupling capacitor, wherein the third funnel-shaped connecting path has an escalating width, the wider part of the third funnel-shaped connecting path is connected to the second resonator, the narrower part of the third funnel-shaped connecting path is connected to the first coupling capacitor; and a fourth funnel-shaped connecting path disposed between the second resonator and the second coupling capacitor, wherein the fourth funnel-shaped connecting path has an escalating width, the wider part of the fourth funnel-shaped connecting path is connected to the second resonator, the narrower part of the fourth funnel-shaped connecting path is connected to the second coupling capacitor; wherein the first funnel-shaped connecting path, the second funnel-shaped connecting path, the third funnel-shaped connecting path and the fourth funnel-shaped connecting path are disposed on the circuit board; the multi-layer

circuit board comprises a first circuit layer, a second circuit layer and a third circuit layer, where the second circuit layer is located between the first circuit layer and the third circuit layer; the first portion of the first resonator is arranged in the first circuit layer, and the second portion of the first resonator is arranged in the third circuit layer; the first portion of the second resonator is arranged in the first circuit layer, and the second portion of the second resonator is arranged in the third circuit layer; the first funnel-shaped connecting path and the third funnel-shaped connecting path are arranged in the first circuit layer; the second funnel-shaped connecting path and the fourth funnel-shaped connecting path are arranged in the third circuit layer; the first coupling capacitor and the first short-circuited stub are arranged in the first circuit layer, and the first coupling capacitor is electrically connected to the first portion of the first resonator and the first portion of the second resonator through the first funnel-shaped connecting path and the third funnel-shaped connecting path; and the second coupling capacitor and the second short-circuited stub are arranged in the third circuit layer, and the second coupling capacitor is electrically connected to the second portion of the first resonator and the second portion of the second resonator through the second funnel-shaped connecting path and the fourth funnel-shaped connecting path.

4. The multi-band filter of claim 3, wherein a number of fingers in each of the interdigital capacitors is 6, and the length of each of the fingers is 140 μm .

5. The multi-band filter of claim 4, wherein the first short-circuited stub and the second short-circuited stub have bending portions.

6. The multi-band filter of claim 4, wherein the first resonator and the second resonator are half-wave resonators.

7. The multi-band filter of claim 4, wherein material of the substrate of the multi-layer circuit board is liquid crystal polymer (LCP), the circuit board is foldable in accordance with a virtual folding line, and the virtual folding line extends passing through the first resonator and the second resonator without passing through the first coupling capacitor and the second coupling capacitor.

8. The multi-band filter of claim 4, wherein the first short-circuited stub and the second short-circuited stub are electrically connected to the grounded layer together.

9. The multi-band filter of claim 1, further comprises: a first funnel-shaped connecting path disposed between the first resonator and the first coupling capacitor, wherein the first funnel-shaped connecting path has an escalating width, the wider part of the first funnel-shaped connecting path is connected to the first resonator, the narrower part of the first funnel-shaped connecting path is connected to the first coupling capacitor; a second funnel-shaped connecting path disposed between the first resonator and the second coupling capacitor, wherein the second funnel-shaped connecting path has an escalating width, the wider part of the second funnel-shaped connecting path is connected to the first resonator, the narrower part of the second funnel-shaped connecting path is connected to the second coupling capacitor; a third funnel-shaped connecting path disposed between the second resonator and the first coupling capacitor, wherein the third funnel-shaped connecting path has an escalating width, the wider part of the third funnel-shaped connecting path is connected to the second resonator, the narrower part of the third funnel-shaped connecting path is connected to the first coupling capacitor; and a fourth funnel-shaped connecting path disposed between the second resonator and the second coupling capacitor, wherein the fourth funnel-shaped connecting path has an escalating width, the wider part of the fourth funnel-shaped connecting path is connected to the second resonator, the narrower part of the fourth funnel-shaped connecting path is connected to the second coupling capacitor; wherein the first funnel-shaped connecting path, the second funnel-shaped connecting path, the third funnel-shaped connecting path and the fourth funnel-shaped connecting path are disposed on the circuit board.

10. The multi-band filter of claim 9, wherein a number of fingers in each of the interdigital capacitors is 6, and the length of each of the fingers is 135 μm .

11. The multi-band filter of claim 10, wherein the first short-circuited stub and the second short-circuited stub have bending portions.

12. The multi-band filter of claim 10, wherein the first resonator and the second resonator are half-wave resonators.

13. The multi-band filter of claim 10, wherein material of the substrate of the circuit board is liquid crystal polymer (LCP), the circuit board is foldable in accordance with a virtual folding line, and the virtual folding line extends passing through the first resonator and the second resonator without passing through the first coupling capacitor and the second coupling capacitor.

14. The multi-band filter of claim 1, wherein the first resonator has a first I/O terminal, the second resonator has a second I/O terminal, the multi-band filter further comprises: a first U-shaped grounded pattern surrounding the first I/O terminal; and a second U-shaped grounded pattern surrounding the second I/O terminal.

15. A multi-band filter, comprising: a circuit board comprising a grounded layer; a first resonator having a first portion and an opposite second portion; a second resonator having a first portion and an opposite second portion; a first interdigital capacitor and a second interdigital capacitor disposed between the first resonator and the second resonator, wherein the first interdigital capacitor and the second interdigital capacitor are configured to couple the first resonator and the second resonator; a first short-circuited stub electrically connected to the first interdigital capacitor; a second short-circuited stub electrically connected to the second interdigital capacitor; and wherein the first resonator, the second resonator, the first interdigital capacitor, the second interdigital capacitor, the first short-circuited stub and the second short-circuited stub are disposed on the circuit board, and the first short-circuited stub and the second short-circuited stub are electrically connected to the grounded layer together through a plurality of vias.
